



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

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SECP3744 – ENTERPRISE SYSTEMS DESIGN AND MODELLING (WBL)

SECTION 1

TITLE:

ENTERPRISE SYSTEM DEVELOPMENT AND MANAGEMENT (ESDM) PROJECT

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CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

Universiti Tun Hussein Onn Malaysia (UTHM) is a public higher education institution that manages diverse academic and administrative activities to support students, academic staff, and university management. These functions are supported by several core operational information systems that collectively form the university's enterprise systems environment. As the scale and complexity of university operations increase, the need for structured coordination between business processes, information systems, and supporting technologies becomes more significant.

Currently, UTHM utilizes a basic form of enterprise architecture to support its information systems landscape. While this provides a general structure for system usage and management, it does not yet represent a comprehensive or formally defined enterprise architecture that fully aligns business needs with application, data, and technology layers. As a result, there is an opportunity to further develop and formalise the enterprise architecture to improve system integration, consistency, and long-term planning.

This project involves the development of a proposed enterprise architecture for UTHM that covers four core operational systems at the enterprise level. Each system is analysed by a different project group to provide detailed architectural insights. The focus of this project is on Student Information System (SIS), with particular emphasis on the Student Course Registration subsystem. By applying enterprise architecture principles commonly used in higher education institutions, this project aims to propose a structured architectural model that supports clearer system alignment, improved information flow, and more effective management of student academic processes at UTHM.

1.2 PROBLEM STATEMENT

This section outlines the key problems that motivate the development of the proposed enterprise architecture for UTHM. It highlights current issues related to the existing enterprise architecture, associated inefficiencies, and potential risks affecting student-related systems. The discussion focuses on the Student Information System (SIS), particularly the Student Course Registration subsystem, to justify the need for a structured enterprise architecture approach.

1.2.1 Current Issues

UTHM utilizes multiple information systems to support its academic and administrative operations, including student related systems such as SIS. However, the enterprise architecture currently applied is basic and not formally documented as a comprehensive enterprise-wide model. As a result, the relationships between business processes, applications, data, and technology are not clearly defined across the university's core operational systems.

1.2.2 Inefficiencies

Due to the absence of a formalized and standardized enterprise architecture, coordination between systems may be inconsistent. System development and enhancement activities may be carried out independently by different units, leading to challenges in system integration and information flow. These inefficiencies are more apparent in student-related systems, where accurate data handling and process alignment are critical to supporting academic activities such as course registration.

1.2.3 Risks and Limitations

Without a clear architectural structure, the Student Course Registration subsystems under the SIS faces limitations in terms of scalability, and long-term planning. The lack of clear mapping between business requirements and system functions increases the risk of misalignment with UTHM broader academic and digital objectives. In addition, future system improvements may become more complex and harder to manage without a structured enterprise architecture to guide decision making.

1.3 PROJECT OBJECTIVES

1.4 PROJECT SCOPE

1.5 PROJECT IMPORTANCE

CHAPTER 2

LITERATURE REVIEW

2.1 INTRODUCTION

2.2 ENTERPRISE ARCHITECTURE

2.3 RELATED SUB-SYSTEM

2.4 TECHNOLOGY USED

2.5 SUMMARY

CHAPTER 3

METHODOLOGY

3.1 INTRODUCTION

This project aims to design an enterprise architecture for Universiti Tun Hussein Onn Malaysia (UTHM). To achieve this goal, the Waterfall development approach is adopted in this project. The methodology guides the development process of this project from starting to the end, which is planning, analysis, design, implementation, and testing the enterprise architecture components. By following the methodology, the project ensures that every phase of the methodology was implemented successfully and aligned with UTHM operational requirements.

3.2 CHOSEN METHODOLOGY

The Waterfall methodology is a sequential development process where every phase must be completed before moving on to the next phase. It is suitable to be applied in projects that have clear and fixed requirements.

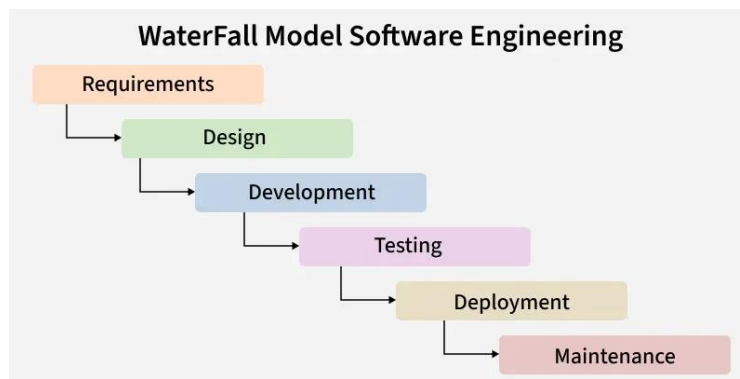


Figure 3.1 Waterfall Methodology Phases (GeeksforGeeks, 2025)

The reason why Waterfall methodology is chosen is because of its structured flow. Since enterprise architecture involves complex systems and long-term planning, understanding the correct requirements is very important. With this methodology, all requirements must be clearly understood before moving to next phase to design the real enterprise architecture.

Other than that, each phase in the Waterfall approach such as requirements analysis, design, implementation, and testing is clearly documented. These documentations will help the university a lot in future maintenance or enhancement.

Moreover, since the phases are clearly defined, it is easier to manage resources, manpower, and project schedules. This reduces uncertainty of the project and reduces the risk of project failure. Therefore, the Waterfall methodology is a suitable approach for this project.

3.3 PHASES OF METHODOLOGY

As illustrated in Figure 3.1, this project is led by the six distinct phases of the Waterfall Model. Each phase depends on the deliverables

3.4 PROJECT PLANNING SCHEDULE

The project planning schedule, as illustrated in Figure xx, follows the sequential stages of the Waterfall methodology to ensure a structure development process. The timeline begins with a collective planning and analysis phase, where the entire team conducts a site visit to UTHM and prepares interview questions to define the system requirements. The project moves into design phase, where individual responsibilities are assigned to all team members to develop specific tasks for the seven modules. The schedule then transitions into the implementation phase, focusing on database creation, and coding. Finally, the process concludes with a testing and deployment phase.

TASKS	ASSIGNED TO	DURATION (WEEKS)	27/10-31/10	3/11-7/11	10/11-14/11	17/11-21/11	1/12-5/12	8/12-12/12	15/12-19/12	22/12-26/12	29/12-2/1	5/1-9/1	12/1-16/1
Meeting with group members	ALL	1											
Project planning	ALL	1											
Requirements													
Describe and Problem Statement	SYAHMI	2											
Project Objectives and Project Scope	NAIM	2											
Project Important	ADRIANA	2											
Literature Review: EA concepts, explore SAP ERP	ALL	2											
Analysis													
Prepare Interview Questions	ALL	1											
Site Visit to UTHM	ALL	1											
Define System Requirements	FARRA	1											
Company Organization Structure	SAFIYA	1											
Analyze Current System	CHUA	1											
Design													
EA AD-ID & TO-BE	ALL	2											
System Architecture	IMAN	2											
Use Case Diagram and Description													
Process 1 : Course List Module	FARRA	1											
Process 2 : Credit Hour Module	NAIM	1											
Process 3 : Course Details Module	ADRIANA	1											
Process 4 : Add Course Module	CHUA	1											
Process 5 : Drop Course Module	IMAN	1											
Process 6 : Section Allocation Module	SYAHMI	1											
Process 7 : Edit Registration Module	SAFIYA	1											
Activity diagram													
Process 1 : Course List Module	FARRA	1											
Process 2 : Credit Hour Module	NAIM	1											
Process 3 : Course Details Module	ADRIANA	1											
Process 4 : Add Course Module	CHUA	1											
Process 5 : Drop Course Module	IMAN	1											
Process 6 : Section Allocation Module	SYAHMI	1											
Process 7 : Edit Registration Module	SAFIYA	1											
Sequence diagram													
Process 1 : Course List Module	FARRA	1											
Process 2 : Credit Hour Module	NAIM	1											
Process 3 : Course Details Module	ADRIANA	1											
Process 4 : Add Course Module	CHUA	1											
Process 5 : Drop Course Module	IMAN	1											
Process 6 : Section Allocation Module	SYAHMI	1											
Process 7 : Edit Registration Module	SAFIYA	1											
Database Design													
Process 1 : Course List Module	FARRA	1											
Process 2 : Credit Hour Module	NAIM	1											
Process 3 : Course Details Module	ADRIANA	1											
Process 4 : Add Course Module	CHUA	1											
Process 5 : Drop Course Module	IMAN	1											
Process 6 : Section Allocation Module	SYAHMI	1											
Process 7 : Edit Registration Module	SAFIYA	1											
Interface Design													
Process 1 : Course List Module	FARRA	1											
Process 2 : Credit Hour Module	NAIM	1											
Process 3 : Course Details Module	ADRIANA	1											
Process 4 : Add Course Module	CHUA	1											
Process 5 : Drop Course Module	IMAN	1											
Process 6 : Section Allocation Module	SYAHMI	1											
Process 7 : Edit Registration Module	SAFIYA	1											
Development													
Create Database													
Process 1 : Course List Module	FARRA	2											
Process 2 : Credit Hour Module	NAIM	2											
Process 3 : Course Details Module	ADRIANA	2											
Process 4 : Add Course Module	CHUA	2											
Process 5 : Drop Course Module	IMAN	2											
Process 6 : Section Allocation Module	SYAHMI	2											
Process 7 : Edit Registration Module	SAFIYA	2											
Coding of system's main function													
Process 1 : Course List Module	FARRA	2											
Process 2 : Credit Hour Module	NAIM	2											
Process 3 : Course Details Module	ADRIANA	2											
Process 4 : Add Course Module	CHUA	2											
Process 5 : Drop Course Module	IMAN	2											
Process 6 : Section Allocation Module	SYAHMI	2											
Process 7 : Edit Registration Module	SAFIYA	2											
Testing & Deployment													
EA Presentation	ALL	1											
Final Project Demo	ALL	1											

Figure : [Gantt Chart](#)

3.5 SUMMARY

CHAPTER 4

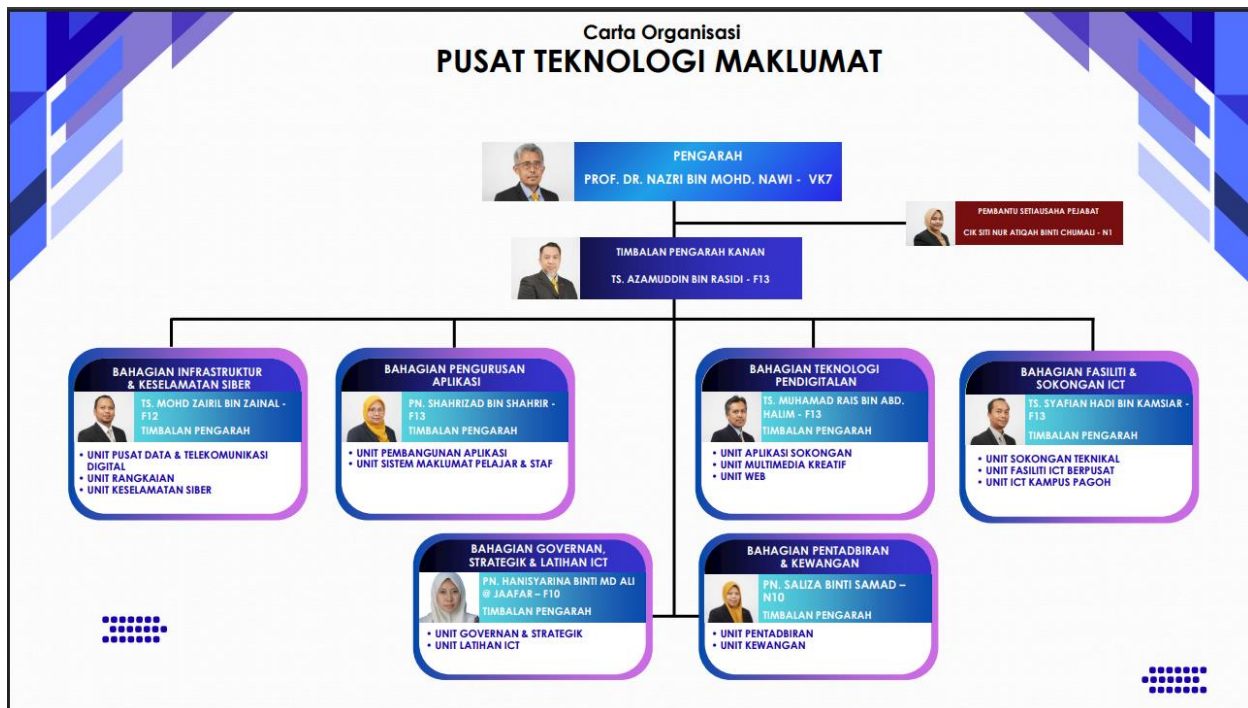
ANALYSIS AND DESIGN

4.1 INTRODUCTION

4.2 COMPANY ORGANIZATION STRUCTURE

The Information Technology Centre (PTM) at Universiti Tun Hussein Onn Malaysia (UTHM) operates under a structured hierarchy led by a Director and a Senior Deputy Director, supported by an Office Secretary Assistant. The department is organized into six main divisions that handle different parts of the university's technology needs. The Application Management Division is responsible for the Application Development Unit and the Student and Staff Information System Unit. Infrastructure and Cyber Security handles data centre, networks, and telecommunications.

The Digitalization Technology, which handles web and multimedia services, and ICT Facilities and Support, which manages technical help and campus equipment. To ensure projects follow university rules, the Governance, Strategic, and ICT Training oversees planning and staff training. Finally, the Administration and Finance division manages the department's budget and office operations.



4.3 CURRENT SYSTEM ANALYSIS

UTHM currently uses a combination of limited manual processes and computerized systems. Their digital operations are supported by a Tier 3 infrastructure. The data center uses Data Structure Infrastructure Management (DCIM) monitoring, air-based condenser cooling system and hot and cold aisle containment to track performance and reduce power consumption. Additionally, they have a staging area where vendors set up systems before entering the containment area.

Furthermore, UTHM's primary digital technologies include a central student database and Learning Management System (LMS) for teaching materials. These systems are not fully integrated so the majority of data transfers are carried out manually one at a time especially when the format or structure varies between the systems. However, API connections are used to manage big data transfers in order to increase efficiency and lower errors. The process was also carried out for data replication to the Data Recovery Center (DRC) where data is transmitted for disaster recovery to a DRC in Pagoh. Moreover, Single Sign-On (SSO) was implemented to make system access easier but frequent user concerns like forgotten password and Wi-Fi login issues still happen.

Additionally, UTHM is transitioning from old systems to a new web-based integrated system called Total Campus Integrated System (TCIS). This migration is ongoing and being completed phase by phase based on modules. Even though UTHM wants to be entirely digital by 2030, some administrative tasks are still done manually because of outdated equipment and legacy systems. They stated that some hardware comes with outdated operating systems which makes it less compatible with modern technologies. In addition, UTHM uses a WAF (Web Application Firewall) to secure web portals and websites as well as two firewalls which filter threats often from China and another from Europe to protect the system and data.

Overall, UTHM is actively pursuing digital transformation and has built a strong infrastructure. However, system integration, manual data handling, legacy equipment and user access continue to be challenges in achieving full end-to-end digital transformation by 2030.

4.4 COMPARISON BETWEEN EXISTING AND PROPOSED SYSTEM

4.5 SYSTEM REQUIREMENTS GATHERING

4.6 SYSTEM REQUIREMENTS

4.7 SYSTEM DESIGN

4.7.1 ENTERPRISE ARCHITECTURE EA – AS IS

4.7.2 ENTERPRISE ARCHITECTURE EA – TO BE

4.7.3 SYSTEM ARCHITECTURE

4.7.4 PROJECT DESIGN

4.7.5 DATABASE DESIGN

4.7.1 INTERFACE DESIGN

4.8 SUMMARY

CHAPTER 5

ANALYSIS AND DESIGN

5.1 INTRODUCTION

5.2 SYSTEM DEVELOPMENT

5.3 CREATE DATABASE

5.4 CODING OF SYSTEM PROCESS

5.5 SUMMARY

CHAPTER 6

CONCLUSION

6.1 INTRODUCTION

6.2 SYSTEM CONTRIBUTION

6.3 SYSTEM CONSTRAINT

6.4 FUTURE SUGGESTION

6.5 SUMMARY